MID SEMESTER EXAMINATION
ODD SEMESTER 2023-2024
MA 202 (PROBABILITY & STATISTICS)
DEPARTMENT OF MATHEMATICAL SCIENCES
INDIAN INSTITUTE OF TECHNOLOGY (BHU), VARANASI-221005

Time:- 2hrs **Total Marks:- 30**

Answer all the questions with proper justification. Stepwise derivation is highly recommended. Start new question from a new page. All notations have their usual meaning.

[1] Let us consider that Urn-I, Urn-II and Urn-III are three identical urns. Urn-I contains 3 white, 2 black, and 5 red balls. Urn-II contains 2 white, 1 black, and 4 red balls. Urn-III contains 3 white, 2 black, and 3 red balls. One urn is chosen at random and 2 balls are drawn. They happen to be red and white. What is the probability that both balls came from Urn-II. (3)

[2] A random variable X has the following distribution

X	-4	-3	-2	-1	0	1	2	3	4
$P(X=x)$	k^2	k	k^2	$2k$	$3k$	$2k$	k	$5k^2$	k^2

Find k and the distribution of $Y=|X-1|+|X+1|$. (3)

[3] Let the random variable X follows Poisson distribution with parameter λ. Derive the function that generates the central moments $E(X-\lambda)^r; r = 1,2,\ldots$. (2)

[4] Let X be a random variable with moment generating function $M_X(t) = \left\{\frac{1}{3}(e^t-1)+1\right\}^3$. Find $P(0 < X \le 2)$. (2)

[5] The random variable X has probability density function $f_X(x) = \frac{1}{2}e^{-|x|}, -\infty < x < \infty$. Find the probability density function of $Y=X^2$. (3)

[6] Let X follow power series distribution with probability mass function

$$P(X=k) = {a_k\theta^k}/{g(\theta)}, \qquad k = 0,1,2,\ldots$$

where $\theta > 0, a_k \ge 0$ and $g(\theta) = \sum_{k=0}^{\infty} a_k\theta^k$. Derive the characteristic function of X. Hence, show that

(i) $E(X) = \theta\frac{g'(\theta)}{g(\theta)}$ and (ii) $Var(X) = \theta\frac{g'(\theta)}{g(\theta)} + \theta^2\left\{\frac{g''(\theta)}{g(\theta)} - \left(\frac{g'(\theta)}{g(\theta)}\right)^2\right\}$. (4)

[7] Find the median and mode (if any) of the continuous distribution with probability density function given by

$$f_X(x) = \begin{cases} k\,(x-9)(10-x), & 9 < x < 10 \\ 0, & otherwise. \end{cases}$$ (3)

[8] The probability density function of a continuous random variable X is given by

$$f_X(x) = \begin{cases} ke^{-b(x-a)}, & a \le x < \infty \\ 0, & elsewhere, \end{cases}$$

where, k, a, b (> 0) are constants. Find k, a, b in terms of μ and σ where μ and σ^2 are mean and variance of the random variable X, respectively. Find the coefficient of skewness. (8)

[9] Let $X \sim N(\mu, \sigma^2)$ and $Z = \frac{X-\mu}{\sigma}$. Then prove with justification that $Z \sim N(0,1)$. (2)

******************************END******************************

Indian Institute of Technology (BHU)
Department of Computer Science and Engineering
Computer Systems Organization (CSO-211)

Time: 2 hours | Mid Term Examination 2023-24 | Max mark: 20

Note: Attempt all questions and make suitable assumption for incomplete information.

1. (a) Design 2-input XOR function using only NAND gates (Note: Minimum number of NAND gates must be used). **(2)**

   (b) Simplify the following Boolean functions using three-variable k-maps
   i) $F(x, y, z) = \sum(3,5,6,7)$
   ii) $F(A, B, C) = \sum(0,2,3,4.6)$ **(2)**

   (c) Simplify the following Boolean functions using four-variable k-maps.
   i) $F(A, B, C, D) = \sum (3, 7, 11, 13, 14, 15)$
   ii) $F(A, B, C, D) = \sum (0, 1, 2, 4, 5, 7, 11, 15)$ **(2)**

2. (a) Design a 2-bit count-down counter. This is a sequential circuit with two flip-flops and one input x. When x = 0, the state of the flip-flops does not change. When x = 1, the state sequence is 11,10,01,00,11, and repeat. **(2)**

   (b) Simplify the following expressions using Boolean algebra.
   i) $AB + A(CD + CD')$
   ii) $(BC' + A'D) (AB' + CD')$ **(2)**

3. (a) Design a combinational circuit with three inputs x, y, z and three outputs A, B, C . When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input **(2.5)**

   (b) Show Given the Boolean function

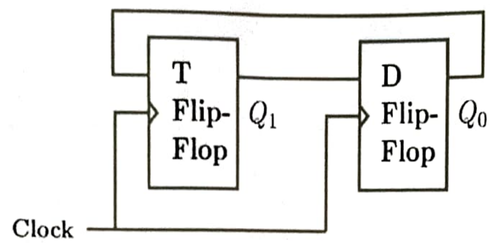
   $F = xy'z + x'y'z + xyz$

   i) List the truth table of the function.

   ii) Draw the logic diagram using the original Boolean expression **(2.5)**

4. (a) A majority function is generated in a combinational circuit when the output is equal to 1 if the input variables have more l's than O's. The output is 0 otherwise. Design a three-input majority function. **(2.5)**

(b) Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop. Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). Show the values for Q_1 and Q_0 for the first 4 cycles. **(2.5)**



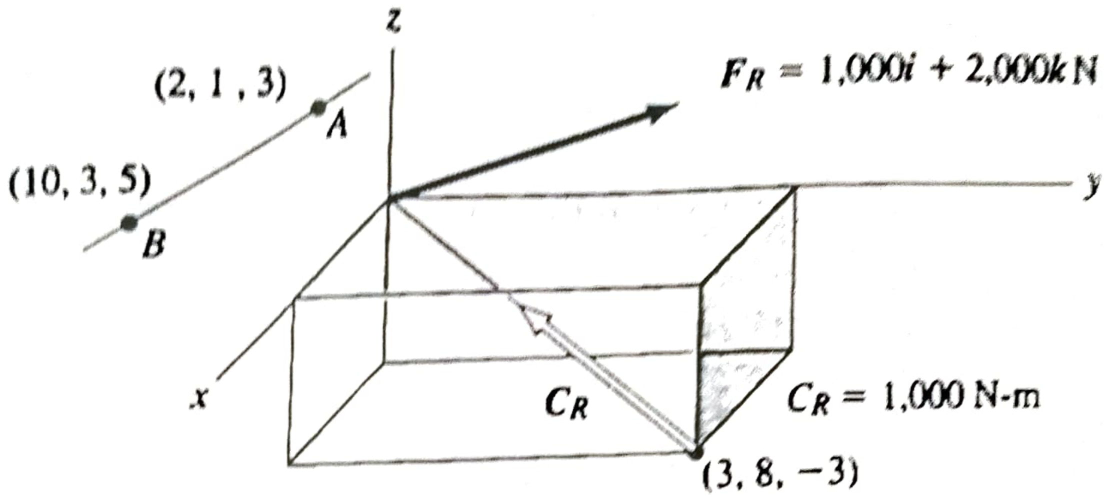
———————— — — — **Best wishes** ———————————— —

Department of Mechanical Engineering,
Indian Institute of Technology, (BHU), Varanasi
ME-102: Engineering Mechanics
Mid Term Examination, Odd Semester 2023–24

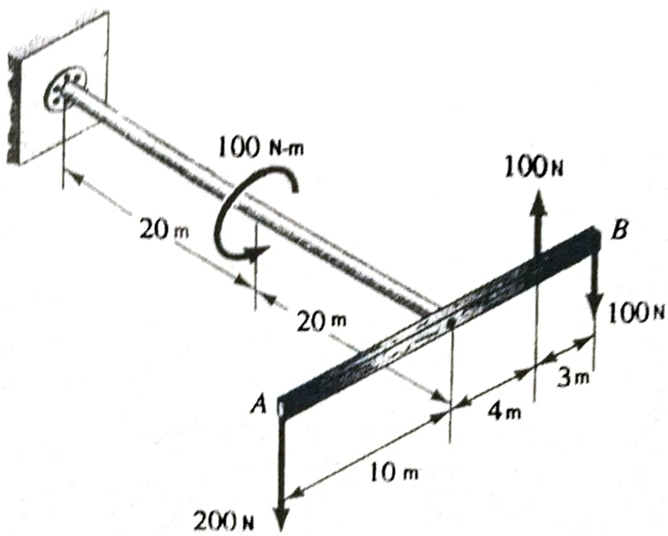
Time: 2 Hours Full Marks: 30

Attempt all the questions.

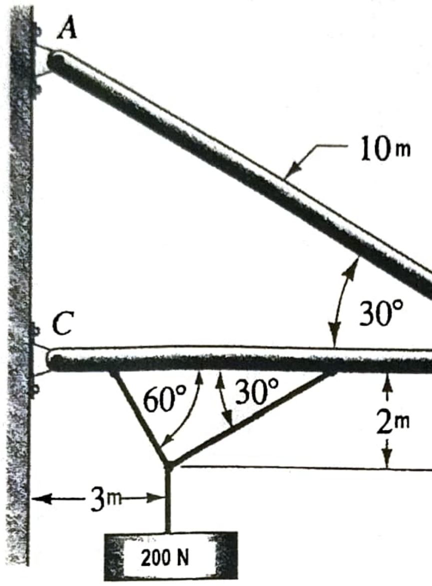
1. Find the ***torque*** about an axis going from point ***A*** to ***B***. [7]



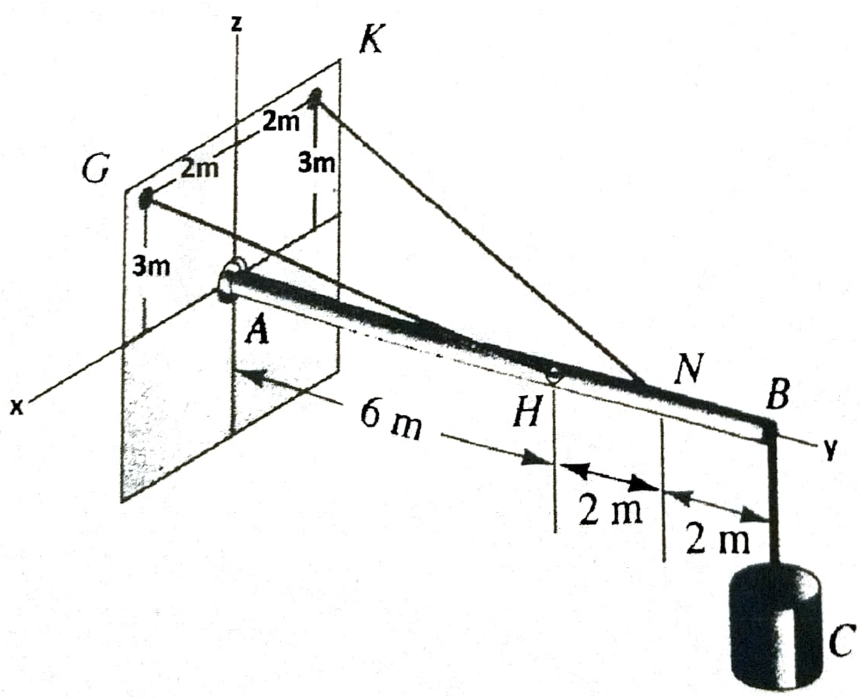
2. Replace the ***forces*** and ***torques*** shown acting on the apparatus by ***a single force***. Carefully give the ***line of action*** of this ***force***. [7]



3. Solve for the ***supporting forces*** at ***A*** and ***C. AB*** weighs ***100N***, and ***BC*** weighs ***150N.*** **[8]**



4. Two cables ***GH*** and ***KN*** support a rod ***AB*** which connects to a ***ball-and-socket*** joint support at ***A*** and supports a ***500kg*** body ***C*** at ***B***. What are the ***tensions*** in the cable and the ***supporting forces*** at ***A***? **[8]**



CSO 204 Discrete Mathematics

Midsemester Examination

All Questions Carry Equal Marks. Solve all the Problems

Max Marks: 80 Time: 2 Hours

1. Recall the definition of a well-formed formula. Show how the following are well-formed formulas using the four rules (How do you get the formulas by application of the rules ?). You may assume P, Q, R are statement variables. $(2.5 \times 4 = 10)$

   (a) $\neg(P \vee Q)$

   (b) $(P \rightarrow (Q \rightarrow R))$

   (c) $(((P \rightarrow Q) \wedge (Q \rightarrow R)) \leftrightarrow (P \rightarrow R))$

   (d) $((P \wedge Q) \leftrightarrow P)$

2. (a) Are equivalences and implications transitive ? Justify your answer. (3)

   (b) Show the following implication $P \rightarrow Q \Rightarrow P \rightarrow (P \wedge Q)$ (4)

   (c) Show the following equivalence $\neg(P \rightarrow Q) \Leftrightarrow P \wedge \neg Q$ (3)

3. (a) Recall the definition of the functionally complete set of connectives. Which subsets of the following set $\{\wedge, \vee, \neg\}$ is functionally complete and is NOT functionally complete? (2)

   (b) Show that $\neg P$ is equivalent to the following formulas $(2 \times 4 = 8)$

   $$\neg\neg\neg P \quad \neg(P \vee (P \wedge Q)) \quad \neg((P \wedge Q) \vee (P \wedge \neg Q)) \quad \neg P \vee (\neg P \wedge \neg Q))$$

4. Obtain the principal disjunctive and conjunctive forms of the following formulas (5+5)

   a. $(Q \rightarrow P) \wedge (\neg P \wedge Q)$

   b. $P \vee (\neg P \rightarrow (Q \vee (\neg Q \rightarrow R)))$

5. (a) Let there be three variables P, Q, R. Let the minterms be $m_0, \ldots, m_7$ and maxterms be $M_0, \ldots, M_7$, What is the minterm m_4 and maxterm M_1 ? (4)

   (b) Write the suffix and prefix form for the following

   $$A \wedge \neg B \qquad A \vee (B \vee C) \qquad A \wedge (B \vee \neg C) \qquad (6)$$

6. (a) What is the difference between sound and valid arguments ? Give an example. (2)

(b) Determine whether the conclusion is valid in the following when $H_1, H_2, \ldots$ are the premises. $(2 \times 4 = 8)$

$H_1 : \neg P$	$H_2 : P \vee Q$	$C : P \wedge Q$
$H_1 : P \to Q$	$H_2 : \neg Q$	$C : P$
$H_1 : R$	$H_2 P \vee \neg P$	$C : R$
$H_1 : P \to (Q \to R)$	$H_2 : P \wedge Q$	$C : R$

7. (a) Show the validity of the conclusion $\neg S$ from the premises $(P \to Q), (\neg Q \vee R) \wedge \neg R, \neg(\neg P \wedge S)$ (5)

(b) Derive the following using rule CP (5)

$$P, P \to (Q \to (R \wedge S)) \Rightarrow Q \to S$$

8. (a) Recall the definition of a sequent. When is a sequent $\alpha \xrightarrow{s} \beta$ is True and False. If a sequent is true is it an axiom schema ? Explain with an example. (4)

(b) Show that the following sets of premises are inconsistent (6)

$$P \to Q, P \to R, Q \to \neg R, P$$
$$A \to (B \to C), D \to (B \wedge \neg C), A \wedge D$$

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INDIAN INSTITUTE OF TECHNOLOGY (BHU), VARANASI - 221 005

Mid-term Examination, Odd Semester 2023-24

H-104: History and Civilization

Max. Marks: 30 **Time: 2 Hours**

Answer any six of the following questions (5 marks each)

1. What are the features of civilization?
2. How do historians construct the historical past? Do you think it is an objective process?
3. What impact did the beginning of agriculture have on the development of human civilization?
4. Why is the Harappan Civilization named so? What are the other names it is known by and why?
5. What are the main features of the Indus Valley Civilization?
6. What do you think is the relation of technology to the history of humankind? Illustrate with some examples.
7. How do you define the prehistoric and the historic period? What are the sources for the study of human past during the prehistoric period?
8. Write a note on technological development in the Stone Age.
9. What is history? Discuss if literature (fiction) can be treated as source for history. Give reasons in support of your argument.